

Engineering LaTeX Templates

A reusable documentation system for reports, technical presentations, and IEEE-style papers.

OPEN-SOURCE ENGINEERING WORKFLOW

LATEX + GIT + PDF

Hossein Molhem

Swipe to explore →

Strong engineering deserves strong documentation.

Technical work often loses impact because the reporting system is rebuilt from scratch for every project.



Repeated setup



Inconsistent structure

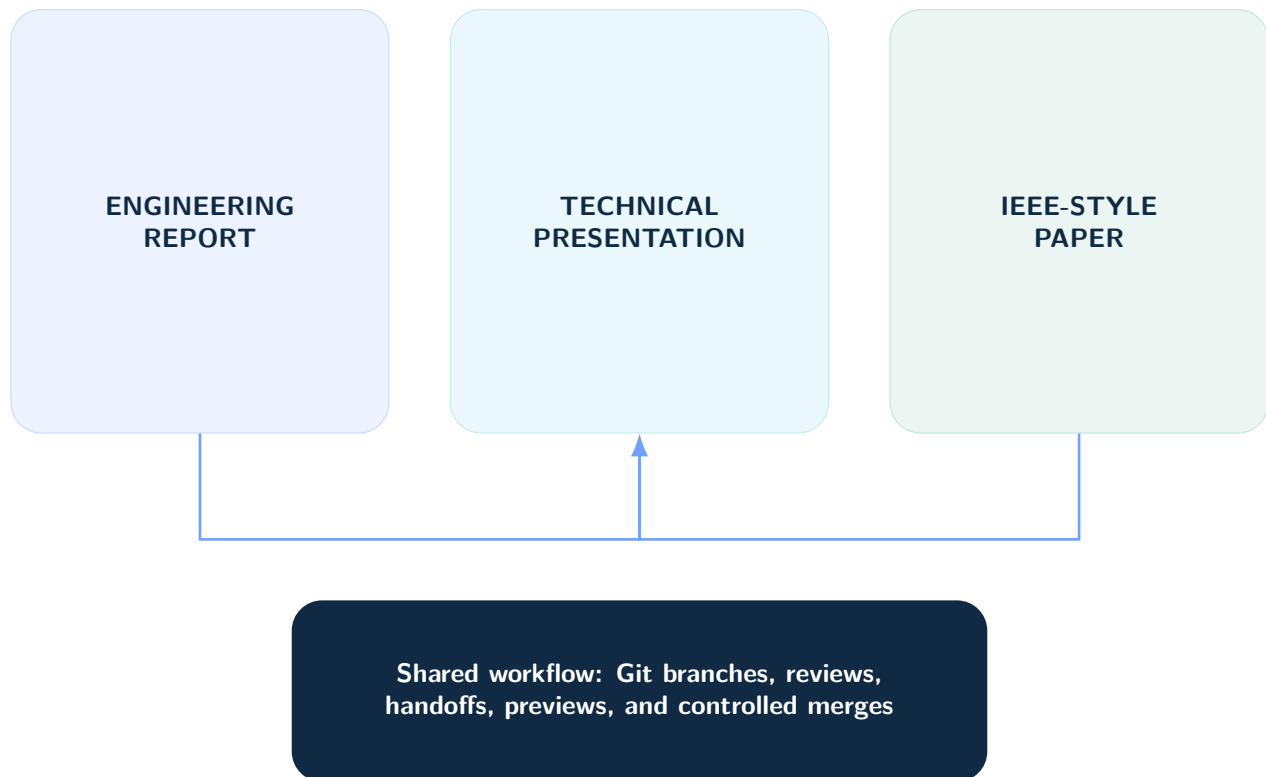


Lost decisions

The core problem: equations, figures, tables, references, appendices, and review decisions are frequently managed as isolated files rather than as one controlled engineering system.

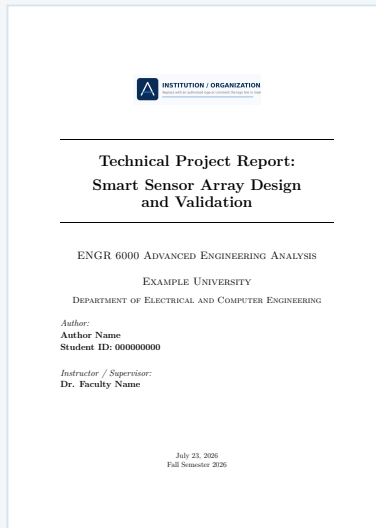
One repository. Three reusable document systems.

Each template provides a practical starting point while preserving consistent engineering structure and version history.

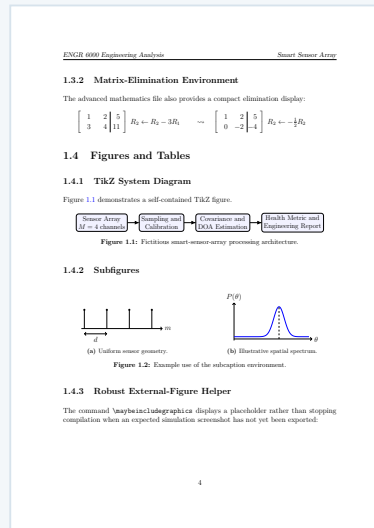


Engineering Report Template

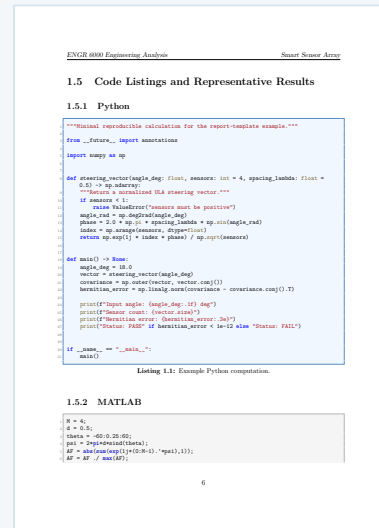
Real pages from the compiled report show the complete path from professional cover design to equations, figures, tables, and source code.



COVER



FIGURES + TABLES



SOURCE CODE

- ✓ Modular front matter and metadata
- ✓ Equations and theorem environments
- ✓ Figures, tables, code, and appendices
- ✓ Repeatable structure for engineering studies

Technical Presentation Template

The compiled deck demonstrates a consistent visual system for opening context, presenting the engineering model, and communicating quantitative results.

Technical Presentation Title
Engineering Analysis, Design, and Results

Hossein Molhem
Institution or Company
Department or Business Unit
July 23, 2026

Hossein MolhemProject Short TitleJuly 23, 20261 / 6

TITLE SYSTEM

Engineering Problem

- Define the physical or computational problem.
- State measurable requirements.
- Identify constraints and assumptions.

Design objective
Replace this block with the primary engineering objective.

$y = \mathbf{H}\mathbf{x} + n$

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SYSTEM MODEL

Key Results

Metric	Baseline	Final
Error	8.2%	1.4%
Runtime	120 ms	24 ms

Engineering interpretation
State what changed, why it changed, and why the result matters.

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RESULTS

LECTURES

DESIGN REVIEWS

RESEARCH UPDATES

ENGINEERING BRIEFINGS

Enhanced IEEE Paper Template

A complete two-page beamforming example proves that the template supports equations, vector figures, tables, citations, and bibliography management.

A Compact Reproducible Example of Conventional and MVDR Beamforming

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Abstract—This paper is a self-contained IEEE-style example intended to demonstrate a complete technical-paper workflow rather than claim a new research contribution. A narrow-band uniform linear array is modeled, a conventional beamformer is compared with a diagonally loaded minimum-variance distortionless-response (MVDR) beamformer, and reproducible synthetic results are reported. The example includes mathematical development, a processing diagram, a publication-quality plot, two tables, cross-references, and a BibTeX bibliography. For the stated eight-element scenario, the illustrative simulation produces an output signal-to-interference-plus-noise ratio of 3.49 dB for conventional beamforming and 15.85 dB for MVDR beamforming.

Index Terms—array signal processing, beamforming, IEEEtran, MVDR, reproducible engineering documentation

I. INTRODUCTION

Array processing combines measurements from spatially distributed sensors to reinforce signals arriving from selected directions while suppressing interference and noise. Conventional beamforming uses a fixed steering vector, whereas adaptive methods use estimated second-order statistics to shape the spatial response [1], [2]. The MVDR formulation is closely related to the high-resolution spatial-spectrum approach introduced by Capon [3].

This paper serves as a reusable IEEE template example. Its technical content is deliberately compact but complete enough to demonstrate equations, figures, tables, citations, labels, and cross-references. The numerical values are generated from a documented synthetic experiment and should be replaced with verified project-specific data before publication.

The principal template features demonstrated here are:

- a standard IEEE conference author block, abstract, and keywords;
- numbered equations with consistent vector and matrix notation;
- an in-document TikZ workflow diagram and an external vector figure;
- compact tables using `booktabs` and `siunitx`; and
- a BibTeX database formatted with the IEEE bibliography style.

II. SIGNAL MODEL

Consider an M -element uniform linear array with inter-element spacing d . The complex baseband snapshot at time

index k is modeled as

$$\mathbf{x}[k] = \mathbf{a}(\theta_s)\mathbf{s}[k] + \mathbf{a}(\theta_i)\mathbf{i}[k] + \mathbf{n}[k], \quad (1)$$

where $\mathbf{s}[k]$ is the desired signal, $\mathbf{i}[k]$ is an interferer, and $\mathbf{n}[k]$ is spatially white noise. For wavelength λ , the steering vector is

$$\mathbf{a}(\theta) = [1 \quad e^{j\mu(\theta)} \quad \dots \quad e^{j(M-1)\mu(\theta)}]^T, \quad (2)$$

with

$$\mu(\theta) = 2\pi \frac{d}{\lambda} \sin \theta. \quad (3)$$

Using K snapshots, the sample covariance matrix is estimated as [4], [5]

$$\hat{\mathbf{R}}_K = \frac{1}{K} \sum_{k=1}^K \mathbf{x}[k]\mathbf{x}^H[k]. \quad (4)$$

III. BEAMFORMER DESIGN

The conventional beamformer steered toward θ_s is

$$\mathbf{w}_{\text{CBF}} = \frac{\mathbf{a}(\theta_s)}{M}. \quad (5)$$

For finite data, a diagonally loaded covariance estimate is used:

$$\hat{\mathbf{R}}_d = \hat{\mathbf{R}}_K + \delta \mathbf{I}, \quad \delta = 10^{-4} \frac{\|\hat{\mathbf{R}}_K\|}{M}. \quad (6)$$

The corresponding MVDR weights are

$$\mathbf{w}_{\text{MVDR}} = \frac{\hat{\mathbf{R}}_d^{-1} \mathbf{a}(\theta_s)}{\mathbf{a}^H(\theta_s) \hat{\mathbf{R}}_d^{-1} \mathbf{a}(\theta_s)}. \quad (7)$$

Equation (7) enforces a distortionless response in the desired direction while minimizing total output power.

The normalized spatial response used for plotting is

$$D(\theta) = 20 \log_{10} \left(\frac{|\mathbf{w}^H \mathbf{a}(\theta)|}{\max_k |\mathbf{w}^H \mathbf{a}(\theta_k)|} \right). \quad (8)$$

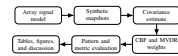


Fig. 1. Reproducible processing and reporting workflow used in the illustrative example.

TABLE I
SYNTHETIC SIMULATION PARAMETERS

Parameter	Value
Array elements, M	8
Element spacing, d (λ)	0.5
Desired direction, θ_s	10°
Interferer direction, θ_i	30°
Snapshots, K	300
Input SNR	10 dB
Input INR	20 dB
Random seed	7

IV. ILLUSTRATIVE EXPERIMENT

The processing sequence is summarized in Fig. 1. A desired source and a single interferer are generated as independent circular complex Gaussian processes. The array receives $K = 300$ snapshots. Table I lists the complete configuration needed to reproduce the example.

The output signal-to-interference-plus-noise ratio is calculated from

$$\text{SINR}_{\text{out}} = \frac{\mathbf{w}^H \mathbf{R}_s \mathbf{w}}{\mathbf{w}^H (\mathbf{R}_i + \mathbf{R}_n) \mathbf{w}}. \quad (9)$$

V. RESULTS AND DISCUSSION

Figure 2 shows that both methods preserve the desired look direction, but MVDR places a substantially deeper response minimum near the interferer. The quantitative values in Table II are produced by the included synthetic configuration rather than copied from an external study.

The MVDR design improves output SINR by approximately 12.36 dB for this scenario. That gain is accompanied by lower white-noise gain, illustrating the practical tradeoff between adaptive interference suppression and robustness. In a research paper, this section should also include sensitivity studies, confidence intervals, computational cost, hardware or measurement uncertainty, and comparisons against credible baselines.

The example also demonstrates why a polished paper needs more than formulas. Equations (1)–(9) define the method, Fig. 1 documents the workflow, Fig. 2 provides visual evidence, and Tables I and II make assumptions and results auditable.

VI. CONCLUSION

This IEEE-style sample demonstrates a reproducible beamforming experiment. The document demonstrates a realistic title block, abstract, citations, cross-referenced equations, two figures, two tables, and a BibTeX bibliography. The technical

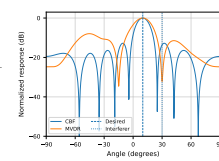


Fig. 2. Normalized conventional and MVDR spatial responses for the synthetic eight-element array. The desired signal is at 10°, and the interferer is at 30°.

TABLE II
ILLUSTRATIVE BEAMFORMER RESULTS

Method	Output SINR (dB)	Response at θ_i (dB)	White-noise gain (dB)
CBF	3.49	-12.63	9.03
MVDR	15.85	-25.03	7.62

example should be treated as illustrative template content. For an actual submission, authors must replace all synthetic values, verify every citation, follow the current instructions of the target IEEE venue, and document the evidence supporting each conclusion.

REFERENCES

- [1] R. D. V. Veen and K. M. Buckley, "Beamforming: A versatile approach to spatial filtering," *IEEE ASSP Magazine*, vol. 5, no. 2, pp. 4–24, Apr. 1989.
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- [3] J. Capon, "High-resolution frequency-wavenumber spectrum analysis," *Proceedings of the IEEE*, vol. 57, no. 8, pp. 1448–1461, Aug. 1969.
- [4] H. Krim and M. Vibert, "Ten decades of array signal processing research: The parametric approach," *IEEE Signal Processing Magazine*, vol. 13, no. 4, pp. 67–96, Jul. 1996.
- [5] P. Stoica and R. L. Moses, *Spectral Analysis of Signals*. Upper Saddle River, NJ, USA: Prentice Hall, 2005.

MODELS + EQUATIONS

FIGURES + TABLES + REFERENCES

Reusable publication workflow: IEEEtran source, separate BibTeX database, cross-referenced equations and figures, and a curated rendered preview.

Documentation treated as an engineering system

Meaningful changes are isolated, reviewed, documented, and merged through a repeatable repository workflow.



Why it matters: the repository records not only the final files, but also implementation scope, validation evidence, limitations, rollback guidance, and follow-up work.

Built for engineering work that needs to be understood, reviewed, and reused.

Academic projects ▪ Research reports ▪ Design reviews
Technical consulting ▪ Reproducible engineering workflows

Explore the repository

github.com/hmolhem/engineering-latex-templates